Pure-Weyl Gravity Programme

Executive summary for physicists

Last substantive update: 16 July 2026.

This is the short, live front door to the programme. It is written for a physicist deciding whether the work intersects their own. It summarizes results; it does not replace the technical manuscripts, certificate ledgers, or the universe-building roadmap.

The sixty-second version

Fourth-order gravity has more local solutions than Einstein gravity, and its extra spin-two branch is normally associated with negative energy or norm. Our starting point is that four logically different questions are often collapsed into that one statement:

1. What modes solve the local field equations?
2. Which modes survive gauge, charge, and boundary reduction?
3. Which reduced structures survive interactions?
4. Which survive quantization and define asymptotic particles?

We have exact answers to substantial parts of the first two questions and sharply scoped results on the third. We do not yet have the fourth.

The most complete field-theory result is for free pure-Weyl gravity on the Lorentzian conformal cylinder, $\mathbb{R} \times S^3$, in a selected closed-universe, zero-charge BV–BFV sector. There the full covariant causal complex is connected to the residual state complex by retarded and advanced chain homotopies. After all fifteen residual $SO(4,2)$ generators are imposed as constraints, the vacuum and one-particle residual sectors are acyclic. The first surviving classes are

$$H^4 = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad G = I_2.$$

These are positive-paired deformation or vertex classes, not positive-norm graviton particles.

The restriction is physical, not cosmetic. On the unrestricted compact linearized phase space, cylinder time translation D carries a charge and is not gauge. It becomes gauge on the full Taub/moment-map zero fibre and its selected derived quotient. A healthy rotating-scalar Berger-cylinder example shows that a matter clock can coexist with a vanishing total D charge at fixed couplings and linear order, but this is not yet a generic, nonlinear, quantum, or asymptotic theorem.

The honest present verdict is therefore:

> The free classical cylinder theorem survives in its declared sector and now has a covariant causal realization. A universal claim that the fourth-order ghost is absent would be false; interactions, quantum anomalies, extra fourth-order branches, and asymptotic scattering remain decisive.

The newest compact Einstein–Maxwell calculation makes the extra-branch qualification concrete. In the generic axial $\ell \geq 2$ block, the Weyl–Maxwell solution module is strictly larger than the Einstein–Maxwell image by two exact polarizations. Their reduced-Hessian Green pairing is nonradical with signature $(2,0)$ in the declared positive-frequency convention. This is not yet a physical norm or particle theorem: the direct four-dimensional action Hessian, complete Lee–Wald matrix, causal boundary conditions, and final quotient are still required.

How claims are typed

Every physical conclusion is indexed by

```
(theory, background, generator, phase space, boundary conditions, lifecycle)
```

We use four plain-language states in this document:

- **Certified**: an exact artifact and an independent verifier establish the scoped claim.

- **Partial**: a real theorem or obstruction is established, but a named dependency needed for the larger conclusion is still open.
- **Fail-closed**: a proposed promotion was tested and is not accepted; this may be a no-go for one ansatz rather than for the theory.
- **Open**: the calculation required for the claim has not been completed.

In particular, `LOCAL-ALGEBRAIC` and `REDUCED-MODE` results are not reported as `LORENTZIAN-CAUSAL` or quantum theorems.

The result spine

Question	Current result	Status	Main limitation
Can the free Pais–Uhlenbeck system admit a positive representation?	The positive symplectic metric is reconstructed canonically and its relation to Krein and real-form choices is classified.	**Certified**	Free representation theory is not interacting stability.
Does that positive structure survive interactions?	Explicit resonant conversion channels obstruct analytic continuation of the free positive metric; Einstein–Weyl cubic order is protected but a physical second-order channel is nonzero.	**Certified / Paper 6 under review**	These are specified interactions and sectors, not a universal no-go for every completion.
What is the selected residual cohomology of free pure-Weyl gravity on $\mathbb{R} \times S^3$?	Vacuum and one-particle sectors are acyclic; $H^4 \cong \mathbb{C}^2$, represented by $[W_+^2]$ and $[W_-^2]$, with normalized Gram matrix I_2.	**Certified; Paper 7 artifact-ready**	All fifteen residual generators, including D, are constrained in a zero-charge closed sector. The classes are deformations, not particles.
Is that residual calculation connected to the covariant field theory?	The complete free metric BV–BFV complex has causal retarded/advanced chain homotopies, compact-to-spacelike-compact transport, residual endpoint recovery, and matching Green/current/residual pairings.	**Certified; Paper 8 artifact-ready**	Conformal cylinder, free classical theory, selected polarization; no Hadamard or quantum construction.
Must D be gauge?	No. It is charged on the unrestricted compact phase space and gauge on the Taub-zero derived sector.	**Certified**	The answer is sector- and boundary-dependent.
Can a healthy clock coexist with total D-gauge?	On one positive Berger-cylinder background, fixed-coupling linearized constraints force $\delta Q_D = 0$ although the matter clock has nonzero internal momentum.	**Certified in the stated linear sector**	Relational observables, nearby backgrounds, nonlinear closure, and quantum stability are not all complete.

Question	Current result	Status	Main limitation
Does the Berger nonlinear Cartan mechanism survive first contact with interactions?	The complete retained q_2 is exact and cyclic; a causal cyclic D-Cartan contraction has been constructed through arity two on all 54 rows.	**Certified algebraically and at the stated classical causal level**	Arity three, QME restoration, and quantum claims are open. The formal source/solution transport algebra is certified, but its distributional microlocal realization remains fail-closed.
Is ordinary Einstein–Maxwell radiation present inside the Weyl–Maxwell system?	The complete standard fixed-bundle harmonic Einstein–Maxwell tangent injects on shell before the final residual quotient, and its Weyl–Maxwell pullback is nondegenerate.	**Certified, reduced-mode/local-algebraic**	The pullback is not generally the Einstein symplectic form; radiative blocks are relatively indefinite, and the complementary branch has only been classified in the generic axial block.
Is the complementary fourth-order axial branch real at the reduced equation level?	For generic compact axial $\ell \geq 2$, the quotient by the Einstein–Maxwell image consists of two exact extra polarizations. Their reduced-Hessian Green pairing is nonradical with signature $(2, 0)$.	**Classified, local-algebraic/reduced-mode**	Direct four-dimensional action-Hessian and Lee–Wald matching, causal boundary admissibility, the final quotient, and particle interpretation are open. The signature is not yet a physical norm.
Is the Einstein sector nonlinearly closed?	Explicit compact fixed-charge photon and gravitational tangents have second-order Taub obstructions; a nonzero null tangent on the universal cover extends in the tested channel.	**Certified examples / classification in progress**	No universal nonlinear Einstein-sector closure theorem.
Can the Berger system support the short-distance structure needed for quantum fields?	The base tensor and ghost wave factors have a certified local Hadamard parametrix, and exact typed Møller intertwiners give the unique formal companion-kernel candidate.	**Partial; microlocal promotion fail-closed**	The order-two transport has no certified Hörmander composition or uniform wavefront control. No companion Hadamard parametrix, global state, QME, or quantum theory is claimed.
Is the quantum theory anomaly-free and unitary?	The even antifield-zero, local dimension-four anomaly candidates reduce to $[\omega C^2]$ and $[\omega E_4]$, with $\omega\Box R$ exact.	**Partial, local-algebraic**	Coefficients, antifield-dependent sectors, QME restoration, Lorentzian time-ordered products, Hadamard state, and asymptotic unitarity are open.

Highlights by audience

Higher-derivative quantization and unitarity

The programme separates the free positive metric, Krein completion, real form, gauge reduction, and interacting deformation problem. The mechanical and flat-field papers show why a free positive representation can be both mathematically canonical and physically insufficient. The gravity papers then ask whether gauge reduction changes the carrier space before positivity is judged.

The potentially useful contribution is a common diagnostic framework for claims that a ghost is removed, rendered null, or made positive. Those claims can disagree because they use different observable algebras, phase spaces, boundary conditions, quotients, or inner products. Our cylinder result does not settle the Mannheim versus Fock–BRST debate; it supplies a third, cohomological reduction whose state type must not be confused with an asymptotic Fock particle.

The key open comparison is one common flat or asymptotic fixture evaluated in all three descriptions.

BV, BRST, and perturbative quantum field theory

Papers 7–8 give an explicit large gauge complex rather than a mode-only quotient. The residual BFV complex is obtained from the metric BV complex, the quadratic moment map is identified with the Taub obstruction, and the derived zero fibre explains why the zero-charge condition cannot be omitted.

For quantum field theorists the present result is a classical input, not a QME theorem. The immediate target is to finish local anomaly cohomology, compute the actual type-A and type-B coefficients by two independent methods, restore or obstruct the QME, and only then ask whether the D-Cartan identity transfers quantum mechanically.

Lorentzian PDE and Green-hyperbolic complexes

The free pure-Weyl metric Hessian does not become a scalar normally hyperbolic operator on a simple same-bundle auxiliary enlargement; exact symbol-rank no-go results explain that failure. The successful object is the whole complex. Curvature prolongation, a symmetric-hyperbolic Weyl–Cotton system, adjoint-tractor transfer, and support-local retracts produce retarded/advanced degree-minus-one homotopies on all 386 prolonged rows.

The externally reusable question is whether this construction can be abstracted into a cyclic transfer theorem for Green-hyperbolic complexes: when a hyperbolic parent complex contracts onto a detour complex, under what hypotheses do causal homotopies and current pairings descend?

On the Berger quantum route, the normally hyperbolic tensor and ghost factors now carry the standard local Hadamard singularity, and the typed source and solution Møller maps satisfy their exact formal intertwining and adjoint identities. The attempted promotion usefully found the remaining analytic obstruction: convergence in finite-slab Sobolev energy norms does not imply that the kernel compositions are defined in the Hörmander sense or that their wavefront cones are uniformly controlled. Thus the formal transported kernel is fixed, while the companion Hadamard claim remains fail-closed.

Conformal geometry, tractors, BGG, and detour complexes

The geometric bridge uses the adjoint-tractor Yang–Mills detour complex and curved BGG/homological transfer to reach the metric Bach complex. This gives Lorentzian causal content to structures usually presented through formal self-adjointness, ellipticity, or representation theory.

The result of interest here is not merely another realization of the Bach operator. It is that the reduced fourth-order operator can inherit causal meaning from a larger tractor complex even when a preferred scalar-principal factorization of the isolated metric operator does not exist. A nearby conformal higher-spin or detour example is the natural portability test.

General relativity and mathematical relativity

The compact Einstein–Maxwell work separates three statements that are often conflated:

1. Einstein solutions solve the Weyl equations on a common background.
2. Einstein linear tangents inject into the Weyl solution space.
3. The Einstein sector is preserved by nonlinear evolution and has the same symplectic form.

The first two now hold in the complete certified standard harmonic tangent before the final quotient. The third does not follow: the Weyl–Maxwell pullback is nondegenerate but differs blockwise from the Einstein–Maxwell form, and fixed-charge second-order Taub obstructions occur for explicit photon and gravitational modes.

The complementary axial block is now explicit rather than inferred from a determinant. For every physical $\ell \geq 2$ in the generic compact domain, the quotient contains two extra solution polarizations. An exact local Green current pairs them nondegenerately and positively at the reduced-Hessian level. That removes a simple radical explanation, but it does not decide the physical sign: the direct action-derived four-dimensional Lee–Wald current and the full Einstein/extra matrix are the next load-bearing comparison.

This connects most directly to linearization stability, charge-fibre obstructions, and the Lorentzian complement to boundary-selected "Einstein from conformal gravity." The decisive next step is the full extra fourth-order quotient and then an asymptotically flat Bach phase space with Bondi/ADM charges and flux.

Relational clocks and the problem of time

The charge calculation replaces the slogan "time is gauge" with a test:

$$\delta H_D = \Omega(\delta\phi, \mathcal{L}_D\phi).$$

For unrestricted compact data this is nonzero. On the Taub-zero sector it vanishes. The Berger construction adds a healthy rotating scalar reference system and shows, at fixed couplings and linear order, that nontrivial matter clock momentum need not make the total D transformation physical.

To connect this to the relational-observable literature, the remaining task is to construct explicit observables $\mathcal{O}_A(\tau)$, prove their gauge invariance and causal dependence, and calculate an actual clock-defined redshift or response.

Amplitudes, twistors, and Einstein from conformal gravity

Our current Einstein result is complementary to boundary selection and twistor embeddings: it asks whether the selected Einstein tangent is a causally and symplectically closed sector, and whether it extends beyond linear order. It does not yet compute an S-matrix.

A useful bridge would identify the certified Einstein projection in helicity or twistor variables and evaluate one cubic or MHV fixture. A later result should determine whether the compact charge-sector obstruction has an amplitude or soft-charge interpretation.

Phenomenology, black holes, and cosmology

This programme is not yet a dark-matter, dark-energy, or black-hole model. Those applications require a physical Lorentzian phase space, stable backgrounds, observables, and boundary charges before fitting rotation curves or expansion histories is meaningful.

The infrastructure is relevant because it can test whether the extra Weyl branch that drives such phenomenology is physical, gauge, constrained, unstable, or boundary-selected. The sequence is: asymptotic and horizon phase spaces; extra-branch classification; nonlinear stability; redshift, lensing, and waveform observables; only then galaxy and cosmology fits.

Where the strongest criticism currently lands

The criticism is correct in three important senses:

- D is not universally gauge. It is charged on the unrestricted compact phase space and is expected to be physical in ordinary asymptotic settings.
- A zero one-particle residual cohomology on the selected closed cylinder is not a proof of particle unitarity or of the absence of radiative degrees of freedom in an asymptotically flat universe.
- The Einstein image is not automatically a nonlinear, symplectic, or exclusive sector of the fourth-order theory.
- In the generic compact axial block, the extra fourth-order module survives the reduced Green-pairing radical test. It can no longer be dismissed there as determinant multiplicity, although its physical norm and admissibility remain undecided.

The selected construction nevertheless survives the criticism where it actually claims to apply:

- the zero-charge condition is now derived and audited rather than assumed silently;
- the residual calculation is connected to the complete covariant causal free complex;
- a healthy clock counterexample shows that matter does not automatically turn total D into a charge;
- the first nonlinear Berger Cartan test closes through arity two, while the still-missing analytic and quantum gates remain explicit.

Thus the sector is neither an arbitrary patch nor a model of the whole universe. It is a mathematically consistent physical choice whose range of stability is now the subject of the programme.

What would materially change the verdict

A strong positive change would be any of:

- a certified Berger relational observable with nontrivial redshift while total D remains gauge;
- arity-three and first resonant-channel survival of the interacting Cartan contraction;
- a direct four-dimensional Lee–Wald match that controls the two extra axial polarizations under causal physical boundary conditions;
- vanishing or removable quantum D-anomaly after QME restoration;
- an asymptotically flat, positive reduced Einstein scattering sector with the extra Weyl branch excluded or controlled.

A strong negative change would be any of:

- failure of the Berger contraction in the next physical interaction channel;

- a direct Lee–Wald calculation that makes the extra axial branch a negative physical direction under admissible boundary conditions;
- an unavoidable nontrivial quantum Cartan anomaly;
- a negative physical extra branch in the asymptotic Bach phase space;
- failure of causal nonlinear closure for every physically admissible Einstein-sector boundary condition.

Either outcome is publishable: the programme is designed to locate the first precise obstruction, not to protect a preferred interpretation.

Reading and verification map

- Series overview and papers
- Paper 7: residual cohomology
- Paper 8: covariant causal transport
- Papers 7–8 computational supplement
- Clean publication-release audit
- Live D-quotient status ledger
- Generic axial extra-branch and pairing report
- Berger typed Møller and microlocal-gate report
- Long-term programme and publication gates
- General-audience introduction

Papers 7–8 are `ARTIFACT_READY`: the manuscripts, supplements, hashes, and clean reproduction audit pass. They are not `SUBMISSION_READY` until human authorship, literature, venue, and prose review and a public archival release are complete.

Update policy

This document should remain short enough to read in roughly ten minutes. A new calculation changes it only when at least one of the following happens:

1. a lifecycle state advances or retreats;
2. a theorem gains or loses a dependency needed for its physical reading;
3. a new background or boundary condition changes a verdict;
4. an open decisive test becomes a certified pass, obstruction, or no-go;
5. a manuscript becomes artifact- or submission-ready.

Raw term counts, implementation milestones, and unverified candidate results belong in team reports, not in this front door. Every update should preserve the strongest limitation immediately beside the headline result.

Changelog

- **16 July 2026:** added the generic axial strict-inclusion theorem and its nonradical reduced Green signature $(2,0)$; recorded the exact typed Møller transport algebra and the fail-closed Hörmander/wavefront gate that prevents promotion to a companion Hadamard parametrix or quantum state.
- **16 July 2026:** created the physicist-facing summary; recorded Papers 7–8 as artifact-ready, the completed cylinder causal bridge, the sector-dependent D verdict, Berger arity-two status with the separate fail-closed analytic import, and the complete standard Einstein–Maxwell harmonic inclusion before the final quotient.